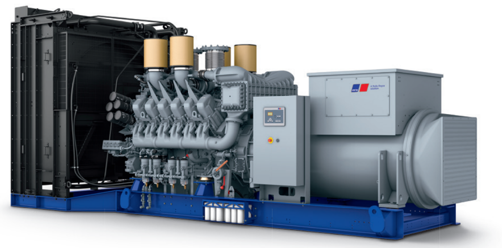
Diesel Generator Set

# mtu 12V4000 DS1750

## 380V – 11 kV/50 Hz/grid stability power/ fuel consumption optimized/12V4000G14F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

## Product highlights

### Benefits
- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

### Support
- Global product support offered

### Standards
- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

### Power rating
- System ratings: 1590 kVA - 1700 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

### Performance assurance certification (PAC)
- Engine-generator set tested to ISO 8528-5 for transient response
- 100% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

### Complete range of accessories available
- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

### Emissions
- Fuel consumption optimized

### Certifications
- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary

A Rolls-Royce solution

## Application data [1]

**Engine**

| | |
|---|---|
| Manufacturer | ***mtu*** |
| Model | 12V4000G14F |
| Type | 4-cycle |
| Arrangement | 12V |
| Displacement: l | 57.2 |
| Bore: mm | 170 |
| Stroke: mm | 210 |
| Compression ratio | 16.4 |
| Rated speed: rpm | 1500 |
| Engine governor | ECU 9 |
| Max power: kWm | 1420 |
| Air cleaner | dry |

**Fuel system**

| | |
|---|---|
| Maximum fuel lift: m | 5 |
| Total fuel flow: l/min | 16 |

| **Fuel consumption [2]** | l/hr | g/kwh |
|---|---|---|
| At 100% of power rating: | 323.3 | 189 |
| At 75% of power rating: | 250.2 | 195 |
| At 50% of power rating: | 173.7 | 203 |

**Liquid capacity (lubrication)**

| | |
|---|---|
| Total oil system capacity: l | 260 |
| Engine jacket water capacity: l | 160 |
| Intercooler coolant capacity: l | 40 |

**Combustion air requirements**

| | |
|---|---|
| Combustion air volume: $m^3/s$ | 1.6 |
| Max. air intake restriction: mbar | 50 |

**Cooling/radiator system**

| | |
|---|---|
| Coolant flow rate (HT circuit): $m^3/hr$ | 56 |
| Coolant flow rate (LT circuit): $m^3/hr$ | 30 |
| Heat rejection to coolant: kW | 540 |
| Heat radiated to charge air cooling: kW | 200 |
| Heat radiated to ambient: kW | 75 |
| Fan power for electr. radiator (40°C): kW | 38 |

**Exhaust system**

| | |
|---|---|
| Exhaust gas temp. (after turbocharger): °C | 430 |
| Exhaust gas volume: $m^3/s$ | 4.0 |
| Maximum allowable back pressure: mbar | 85 |
| Minimum allowable back pressure: mbar | 30 |

## Standard and optional features

**System ratings (kW/kVA)**

| Generator model | Voltage | Fuel consumption optimized | | | | | |
|---|---|---|---|---|---|---|---|
| | | without radiator | | | with mechanical radiator | | |
| | | kWel | kVA* | AMPS | kWel | kVA* | AMPS |
| Leroy Somer LSA52.3 S5 (Low voltage Leroy Somer standard) | 380 V | 1360 | 1700 | 2583 | 1320 | 1650 | 2507 |
| | 400 V | 1360 | 1700 | 2454 | 1320 | 1650 | 2382 |
| | 415 V | 1360 | 1700 | 2365 | 1320 | 1650 | 2295 |
| Marathon 743RSL7090 (Low voltage Marathon) | 380 V | 1352 | 1690 | 2568 | 1312 | 1640 | 2492 |
| | 400 V | 1344 | 1680 | 2425 | 1312 | 1640 | 2367 |
| | 415 V | 1272 | 1590 | 2212 | 1272 | 1590 | 2212 |
| Marathon 744RSL7091 (Low voltage Marathon oversized) | 380 V | 1352 | 1690 | 2568 | 1312 | 1640 | 2492 |
| | 400 V | 1344 | 1680 | 2425 | 1312 | 1640 | 2367 |
| | 415 V | 1272 | 1590 | 2212 | 1272 | 1590 | 2212 |
| Marathon 1020FDH7095 (Medium volt. marathon) | 11 kV | 1352 | 1690 | 89 | 1312 | 1640 | 86 |
| Leroy Somer LSA53.2 VL6 (Medium volt. Leroy Somer) | 11 kV | 1352 | 1690 | 89 | 1320 | 1650 | 87 |

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).

2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.